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Section 1: User Instructions

1.0 Code

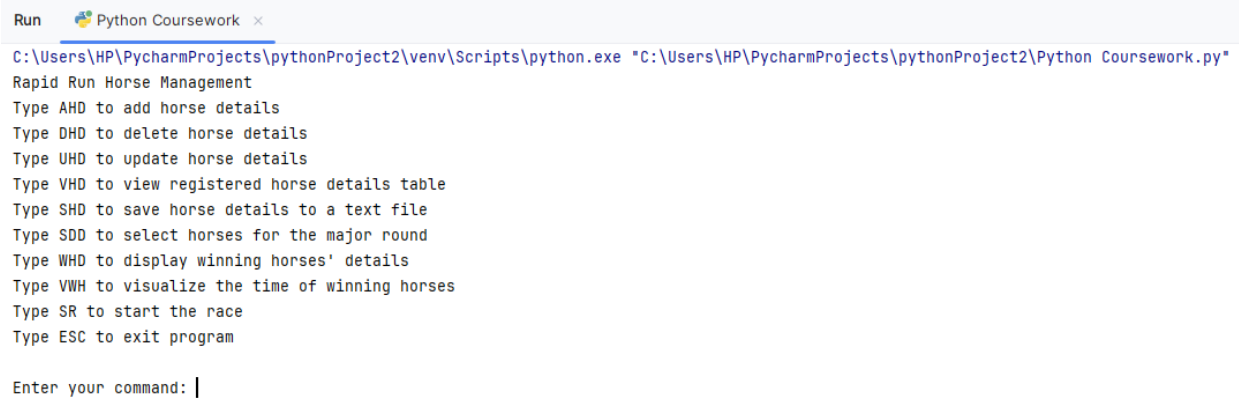
```
Python Coursework.py x
302
303 # Instructions for the user
304 1 usage
305 def rapid_run():
306     while True:
307         print("Rapid Run Horse Management\n"
308               "Type AHD to add horse details\n"
309               "Type DHD to delete horse details\n"
310               "Type UHD to update horse details\n"
311               "Type VHD to view registered horse details table\n"
312               "Type SHD to save horse details to a text file\n"
313               "Type SDD to select horses for the major round\n"
314               "Type WHD to display winning horses' details\n"
315               "Type VWH to visualize the time of winning horses\n"
316               "Type SR to start the race\n"
317               "Type ESC to exit program\n")
318         command = input("Enter your command: ").upper()
```

Figure 1.1: Instructions for the user Code

```
Python Coursework.py x
320
321 if command == "AHD":
322     add_horse_details()
323 elif command == "DHD":
324     delete_horse_details()
325 elif command == "UHD":
326     update_horse_details()
327 elif command == "VHD":
328     view_sorted_list()
329 elif command == "SHD":
330     save_to_text_file()
331 elif command == "SDD":
332     selection_for_major_round()
333 elif command == "WHD":
334     display_winners()
335 elif command == "VWH":
336     visualizing_winner(race_times)
337 elif command == "SR":
338     starting_status_of_the_race()
339 elif command == "ESC":
340     print("Exit")
341     break
342 else:
343     print("Please enter a valid command!")
344
345 rapid_run()
```

Figure 1.2: Instructions for the user Code

1.1 Result



```
Run Python Coursework x
C:\Users\HP\PycharmProjects\pythonProject2\venv\Scripts\python.exe "C:\Users\HP\PycharmProjects\pythonProject2\Python Coursework.py"
Rapid Run Horse Management
Type AHD to add horse details
Type DHD to delete horse details
Type UHD to update horse details
Type VHD to view registered horse details table
Type SHD to save horse details to a text file
Type SDD to select horses for the major round
Type WHD to display winning horses' details
Type VWH to visualize the time of winning horses
Type SR to start the race
Type ESC to exit program

Enter your command: |
```

Figure 1.3: Instructions for the user Result

1.2 Code Description

Allows the user to input the function that needs to be executed.

This is an infinite loop which shows the command needs to be input for each function.

After the user input the command it checks whether it is a valid command and executes the function related to the command.

If the user input is invalid it prints an error message.

The program terminates when the user input "ESC" command.

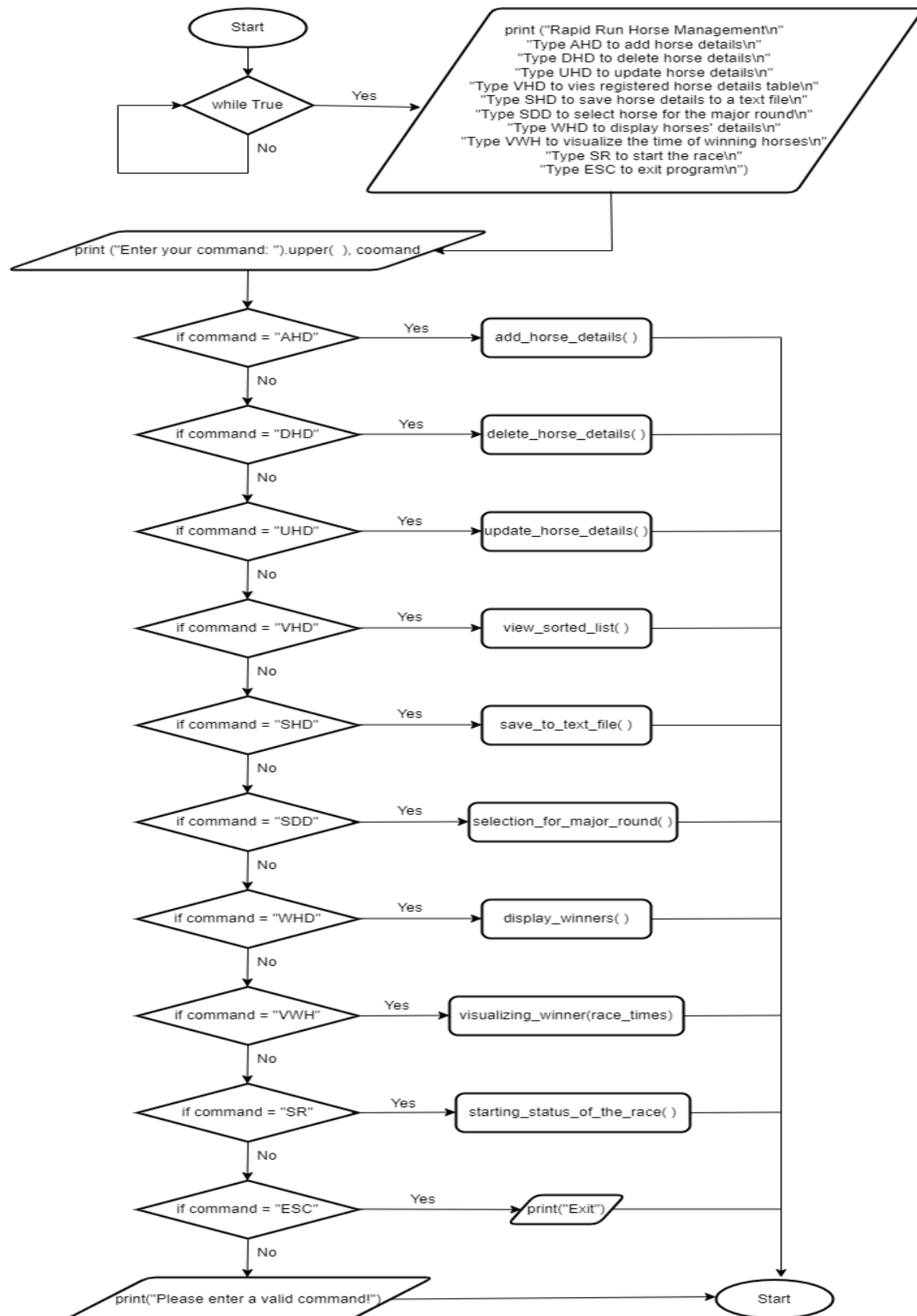


Figure 1.4: Instructions for the user Flow Chart

Section 2: Start Race

1.0 Code

A screenshot of a code editor window titled 'Python Coursework.py'. The code defines a function 'starting_status_of_the_race()' which checks if a race has started. It uses a global variable 'Race_has_started'. If it's not started, it prints 'Race starting!!' and sets the variable to True. Otherwise, it prints 'Race has already started!'.

```
292  
293 # Function to start the race  
    1 usage  
294 def starting_status_of_the_race():  
295     global Race_has_started  
296     if not Race_has_started:  
297         print("Race starting!!\n")  
298         Race_has_started = True  
299     else:  
300         print("Race has already started!\n")
```

Figure 2.1: Starting the race Code

2.1 Result

```
Enter your command: SR  
Race starting!!
```

Figure 2.2: Starting the race Result

2.2 Code Description

Allows user to start the race after entering the details of the horses.

If the user starts the race as the first command, user will not be able to enter, delete, or update horse details.

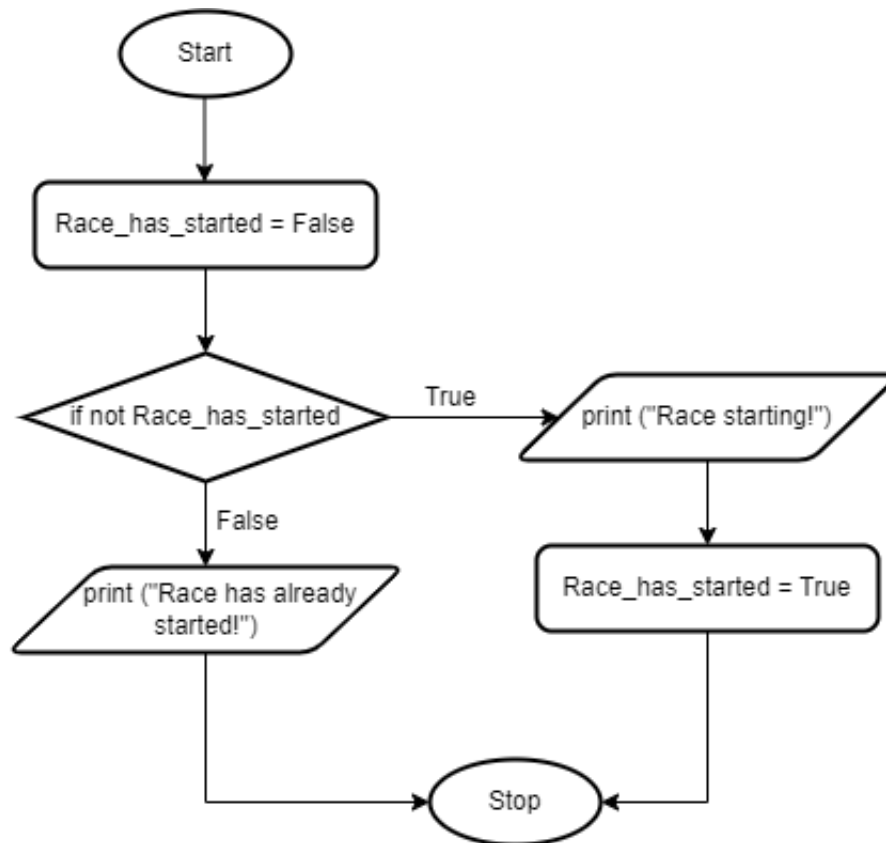


Figure 2.3: Starting the race Flow Chart

Section 3: Add Horse Details

3.0 Code

```
Python Coursework.py ×
1 import random
2
3 # Creating a list to store horse details
4 file_path = 'horse_details.txt' # Assuming this is the path to the file
5 GROUPS = ['A', 'B', 'C', 'D']
6 horse_list = []
7 selected_horses = []
8 race_times = {}
9 Race_has_started = False
```

Figure 3.1: Adding horse details Code

```

Python Coursework.py x
13 # Adding horse details
14 1 usage
15 def add_horse_details():
16     global horse_list, Race_has_started
17     if not Race_has_started:
18         while True:
19             try:
20                 horse_id = int(input("Enter Horse ID: "))
21                 break
22             except ValueError:
23                 print("Enter a valid integer value for Horse ID")
24     horse_name = input("Enter Horse Name: ")
25     jockey_name = input("Enter Jockey Name: ")
26     while True:
27         try:
28             age = int(input("Enter Age: "))
29             break
30         except ValueError:
31             print("Enter a valid positive integer value for age greater than 0")
32     breed = input("Enter Breed of the Horse: ")
33     race_record = input("Enter Race Record of the Horse: ")
34     while True:
35         group_list = ['A', 'B', 'C', 'D']
36         group = input("Enter the Group: ").upper()
37         if group not in group_list:
38             print("Enter a valid group out of", group_list)

```

Figure 3.2: Adding horse details Code

```

Python Coursework.py x
36     if group not in group_list:
37         print("Enter a valid group out of", group_list)
38     else:
39         break
40
41     '#Creating a dictionary for the newly added horse details'
42     horse_details = {"Horse ID": horse_id,
43                     "Horse Name": horse_name,
44                     "Jockey Name": jockey_name,
45                     "Age": age,
46                     "Breed": breed,
47                     "Race Record": race_record,
48                     "Group": group}
49
50     horse_list.append(horse_details)
51     print("Your horse details are added to the list successfully!\n", horse_list)
52 else:
53     print("Race has already started!")
54
55

```

Figure 3.3: Adding horse details Code

3.1 Result

For one horse detail:

```
Enter your command: AHD
Enter Horse ID: 1
Enter Horse Name: Maximus
Enter Jockey Name: David Thompson
Enter Age: 10
Enter Breed of the Horse: Arabian
Enter Race Record of the Horse: 8 wins in 10 races
Enter the Group: A
Your horse details are added to the list successfully!
[{'Horse ID': 1, 'Horse Name': 'Maximus', 'Jockey Name': 'David Thompson', 'Age': 10, 'Breed': 'Arabian', 'Race Record': '8 wins in 10 races', 'Group': 'A'}]
```

Figure 3.4: Adding horse details Result

Completed list created after adding all 20 horse details:

```
Your horse details are added to the list successfully!
[{'Horse ID': 1, 'Horse Name': 'Maximus', 'Jockey Name': 'David Thompson', 'Age': 10, 'Breed': 'Arabian', 'Race Record': '8 wins in 10 races', 'Group': 'A'}, {'Horse ID': 2, 'Horse Name': 'Ruby', 'Jockey Name': 'Sally Kendall', 'Age': 12, 'Breed': 'Thoroughbred', 'Race Record': '9 wins in 11 races', 'Group': 'A'}, {'Horse ID': 3, 'Horse Name': 'Ranger', 'Jockey Name': 'John Maxwell', 'Age': 8, 'Breed': 'Appaloosa', 'Race Record': '4 wins in 7 races', 'Group': 'A'}, {'Horse ID': 4, 'Horse Name': 'Molly', 'Jockey Name': 'Harry Clanden', 'Age': 13, 'Breed': 'Morgan', 'Race Record': '7 wins in 11 races', 'Group': 'A'}, {'Horse ID': 5, 'Horse Name': 'Joey', 'Jockey Name': 'Belle Kendall', 'Age': 13, 'Breed': 'Warmbloods', 'Race Record': '8 wins in 12 races', 'Group': 'A'}, {'Horse ID': 6, 'Horse Name': 'Fury', 'Jockey Name': 'Daniel Baker', 'Age': 8, 'Breed': 'Gaited Breeds', 'Race Record': '3 wins in 8 races', 'Group': 'B'}, {'Horse ID': 7, 'Horse Name': 'Lucky', 'Jockey Name': 'Jonathan Cooper', 'Age': 15, 'Breed': 'Draft Breeds', 'Race Record': '13 wins in 14 races', 'Group': 'B'}, {'Horse ID': 8, 'Horse Name': 'Dakota', 'Jockey Name': 'Ellie Morgan', 'Age': 8, 'Breed': 'American Quarter', 'Race Record': '5 wins in 10 races', 'Group': 'B'}, {'Horse ID': 9, 'Horse Name': 'Chad', 'Jockey Name': 'Edward Matthew', 'Age': 10, 'Breed': 'Appaloosa', 'Race Record': '7 wins in 12 races', 'Group': 'B'}, {'Horse ID': 10, 'Horse Name': 'Dusty', 'Jockey Name': 'Henry Kelson', 'Age': 13, 'Breed': 'Arabian', 'Race Record': '10 wins in 14 races', 'Group': 'B'}, {'Horse ID': 11, 'Horse Name': 'Sparky', 'Jockey Name': 'Peter Edinson', 'Age': 10, 'Breed': 'Thoroughbred', 'Race Record': '7 wins in 10 races', 'Group': 'C'}, {'Horse ID': 12, 'Horse Name': 'Mocha', 'Jockey Name': 'Emma Baker', 'Age': 13, 'Breed': 'Warmbloods', 'Race Record': '9 wins in 15 races', 'Group': 'D'}, {'Horse ID': 13, 'Horse Name': 'Nutmeg', 'Jockey Name': 'Jack Davis', 'Age': 9, 'Breed': 'Gaited Breeds', 'Race Record': '6 wins in 9 races', 'Group': 'C'}, {'Horse ID': 14, 'Horse Name': 'Trixie', 'Jockey Name': 'Helen Ashley', 'Age': 11, 'Breed': 'American Quarter', 'Race Record': '6 wins in 10 races', 'Group': 'D'}, {'Horse ID': 15, 'Horse Name': 'Jackie', 'Jockey Name': 'Louis McKale', 'Age': 10, 'Breed': 'Arabian', 'Race Record': '5 wins in 9 races', 'Group': 'C'}, {'Horse ID': 16, 'Horse Name': 'Hidalgo', 'Jockey Name': 'Carl Peterson', 'Age': 7, 'Breed': 'Draft Breeds', 'Race Record': '3 wins in 5 races', 'Group': 'D'}, {'Horse ID': 17, 'Horse Name': 'Charlie', 'Jockey Name': 'Sara Lennon', 'Age': 10, 'Breed': 'Morgan', 'Race Record': '6 wins in 11 races', 'Group': 'C'}, {'Horse ID': 18, 'Horse Name': 'Billy', 'Jockey Name': 'Hardin Wellington', 'Age': 10, 'Breed': 'Arabian', 'Race Record': '8 wins in 9 races', 'Group': 'D'}, {'Horse ID': 19, 'Horse Name': 'Dash', 'Jockey Name': 'Nick Marley', 'Age': 13, 'Breed': 'Appaloosa', 'Race Record': '10 wins in 13 races', 'Group': 'C'}, {'Horse ID': 20, 'Horse Name': 'Ollie', 'Jockey Name': 'Kate Parker', 'Age': 12, 'Breed': 'American Quarter', 'Race Record': '9 wins in 13 races', 'Group': 'D'}]
```

Figure 3.5: Adding horse details Result

3.2 Code Description

Allows the user to enter the details of the horses. This code will not execute if the race has already started. If the user enters an invalid value for “Horse ID”, “Age”, and “Group”, an error message will be raised showing the related error and give another chance to enter a valid value. A dictionary (“horse_details”) will be created to store the horse details and the dictionary will be added to the list (“horse_lits”) at the end of the data inserting.

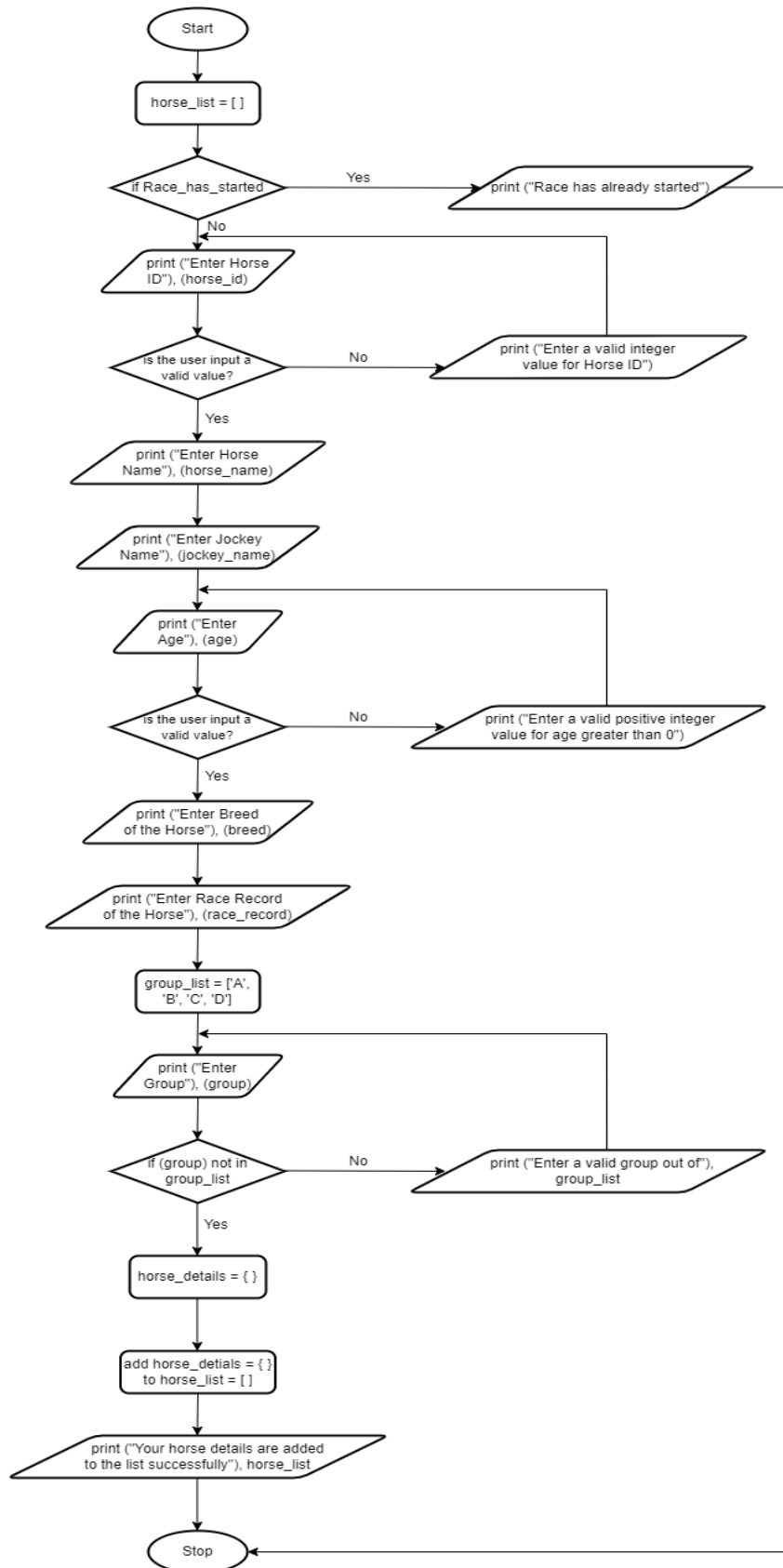


Figure 3.6: Adding horse details Flow Chart

Section 4: Delete Horse Details

4.0 Code

```
Python Coursework.py x
56 # Deleting horse details
57 1 usage
58 def delete_horse_details():
59     global horse_list, Race_has_started
60     if not Race_has_started:
61         if not horse_list:
62             print("Cannot delete record as a list of horses doesn't exist")
63         else:
64             id_horse = int(input("Enter Horse ID needed to be deleted: "))
65             for h in horse_list:
66                 if h["Horse ID"] == id_horse:
67                     horse_list.remove(h)
68                     print("Horse details of the horse ID were successfully removed!\n", horse_list)
69                 else:
70                     print("Horse ID was not found!")
71     else:
72         print("Race has already started!")
73
```

Figure 4.1: Deleting horse details Code

4.1 Result

```
Enter your command: DHD
Enter Horse ID needed to be deleted: 21
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse ID was not found!
Horse details of the horse ID were successfully removed!
[{'Horse ID': 1, 'Horse Name': 'Maximus', 'Jockey Name': 'David Thompson', 'Age': 10, 'Breed': 'Arabian', 'Race Record': '8 wins in 10 races', 'Group': 'A'}, {'Horse ID': 2, 'Horse Name': 'Ruhv', 'Jockey Name': 'Sally Kendall', 'Age': 12, 'Breed': 'Thoroughbred', 'Race Record': '9 wins in 11 races'}
```

Figure 4.2: Deleting horse details Result

4.2 Code Description

Allows the user to delete unwanted records out of the horse details. If the race has already started it will print an error message mentioning the race has started and cannot delete records while mentioning cannot delete records as a list does not exist if the "horse_list" has not been created. After the user input the "Horse ID" of the record required to be deleted, it checks through all the "Horse ID"s and removes the related record using the for loop. If the user input cannot be found, it will print that the "Horse ID was not found".

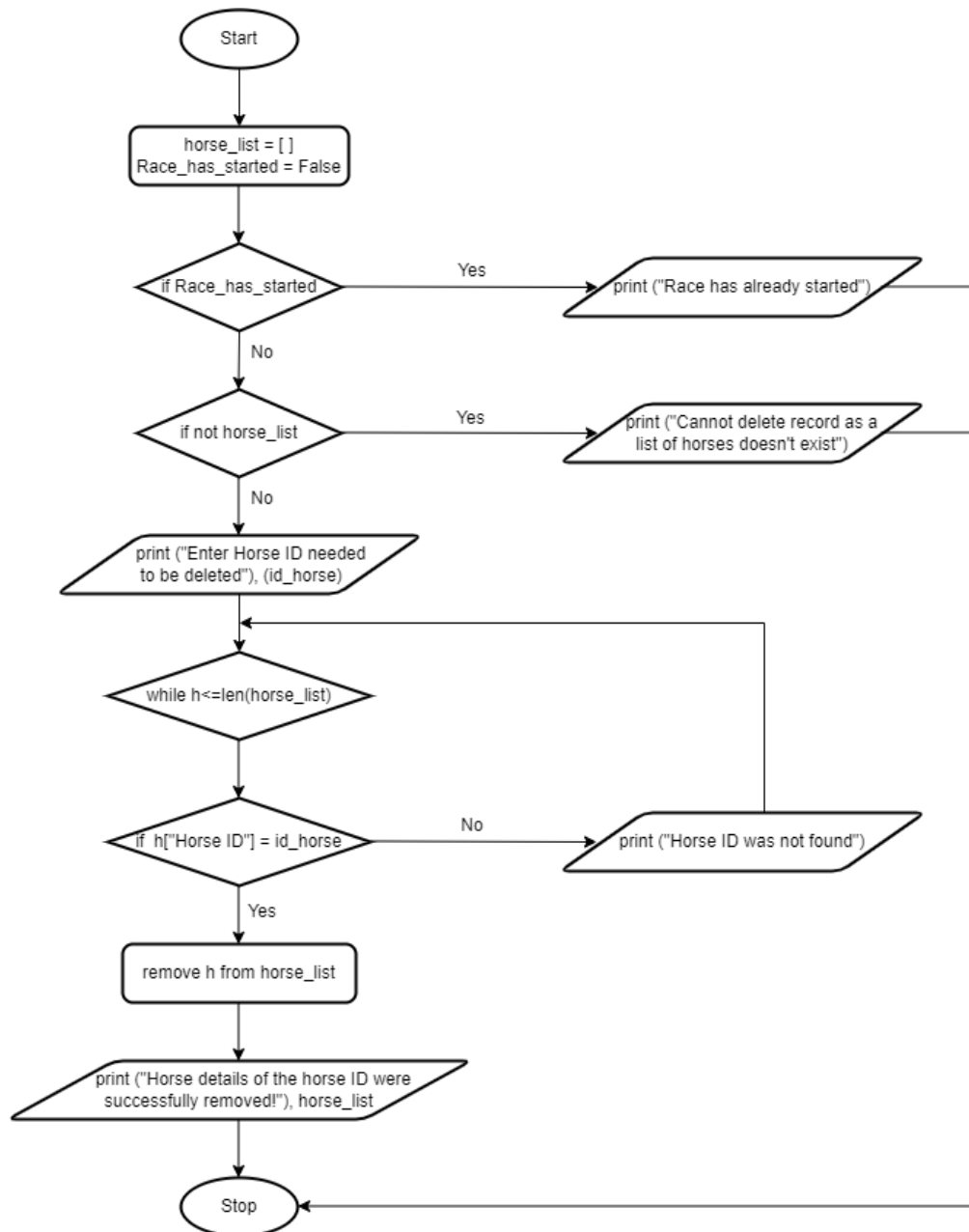


Figure 4.3: Deleting horse details Flow Chart

Section 5: Update Horse Details

5.0 Code

Python Coursework.py ×

```
75 # Updating horse details
76 1 usage
77 def update_horse_details():
78     global horse_list, Race_has_started
79     if not Race_has_started:
80         if not horse_list:
81             print("Cannot update record as a list of horses doesn't exist")
82         else:
83             id_horse = int(input("Enter Horse ID needed to be updated: "))
84
85             for h in horse_list:
86                 if h["Horse ID"] == id_horse:
87                     print("Enter the details of the horse needed to be updated with horse ID = ", id_horse)
88                     while True:
89                         try:
90                             h["Horse ID"] = int(input("Enter updated Horse ID: "))
91                             break
92                         except ValueError:
93                             print("Enter a valid integer value for the updated Horse ID")
94                     h["Horse Name"] = input("Enter updated Horse Name: ")
95                     h["Jockey Name"] = input("Enter updated Jockey Name: ")
96                     while True:
97                         try:
98                             h["Age"] = int(input("Enter updated Age: "))
99                             break
100                         except ValueError:
101                             print("Enter a valid positive integer value for age greater than 0")
```

Figure 5.1: Updating horse details Code

Python Coursework.py ×

```
99     except ValueError:
100         print("Enter a valid positive integer value for age greater than 0")
101     h["Breed"] = input("Enter updated Breed of the Horse: ")
102     h["Race Record"] = input("Enter updated Race Record of the Horse: ")
103     while True:
104         group_list = ['A', 'B', 'C', 'D']
105         h["Group"] = input("Enter the updated Group: ").upper()
106         if h["Group"] not in group_list:
107             print("Enter a valid group out of", group_list)
108         else:
109             break
110     print("Your new horse details are updated successfully!\n", horse_list)
111 else:
112     print("Horse ID was not found!")
113 else:
114     print("Race has already started!")
115
```

Figure 5.2: Updating horse details Code

5.1 Result

```
Enter your command: UHD  
Enter Horse ID needed to be updated: 20  
Horse ID was not found!  
Horse ID was not found!  
Horse ID was not found!  
Horse ID was not found!  
Horse ID was not found!  
Horse ID was not found!  
Horse ID was not found!  
Horse ID was not found!  
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Horse ID was not found!  
Horse ID was not found!  
Horse ID was not found!  
Horse ID was not found!  
Horse ID was not found!  
Horse ID was not found!  
  
Enter the details of the horse needed to be updated with horse ID = 20  
Enter updated Horse ID: 20  
Enter updated Horse Name: Ollie
```

Figure 5.3: Updating horse details Result

```
Enter updated Horse Name: Ollie
Enter updated Jockey Name: Kate Huston
Enter updated Age: 14
Enter updated Breed of the Horse: American Quarter
Enter updated Race Record of the Horse: 11 wins in 15 races
Enter the updated Group: D
Your new horse details are updated successfully!
```

Figure 5.4: Updating horse details Result

Completed list created after updating horse details:

Your new horse details are updated successfully!

```
[{'Horse ID': 1, 'Horse Name': 'Maximus', 'Jockey Name': 'David Thompson', 'Age': 10, 'Breed': 'Arabian', 'Race Record': '8 wins in 10 races', 'Group': 'A'}, {'Horse ID': 2, 'Horse Name': 'Ruby', 'Jockey Name': 'Sally Kendall', 'Age': 12, 'Breed': 'Thoroughbred', 'Race Record': '9 wins in 11 races', 'Group': 'A'}, {'Horse ID': 3, 'Horse Name': 'Ranger', 'Jockey Name': 'John Maxwell', 'Age': 8, 'Breed': 'Appaloosa', 'Race Record': '4 wins in 7 races', 'Group': 'A'}, {'Horse ID': 4, 'Horse Name': 'Molly', 'Jockey Name': 'Harry Clanden', 'Age': 13, 'Breed': 'Morgan', 'Race Record': '7 wins in 11 races', 'Group': 'A'}, {'Horse ID': 5, 'Horse Name': 'Joey', 'Jockey Name': 'Belle Kendall', 'Age': 13, 'Breed': 'Warmbloods', 'Race Record': '8 wins in 12 races', 'Group': 'A'}, {'Horse ID': 6, 'Horse Name': 'Fury', 'Jockey Name': 'Daniel Baker', 'Age': 8, 'Breed': 'Gaited Breeds', 'Race Record': '3 wins in 8 races', 'Group': 'B'}, {'Horse ID': 7, 'Horse Name': 'Lucky', 'Jockey Name': 'Jonathan Cooper', 'Age': 15, 'Breed': 'Draft Breeds', 'Race Record': '13 wins in 14 races', 'Group': 'B'}, {'Horse ID': 8, 'Horse Name': 'Dakota', 'Jockey Name': 'Ellie Morgan', 'Age': 8, 'Breed': 'American Quarter', 'Race Record': '5 wins in 10 races', 'Group': 'B'}, {'Horse ID': 9, 'Horse Name': 'Chad', 'Jockey Name': 'Edward Matthew', 'Age': 10, 'Breed': 'Appaloosa', 'Race Record': '7 wins in 12 races', 'Group': 'B'}, {'Horse ID': 10, 'Horse Name': 'Dusty', 'Jockey Name': 'Henry Kelson', 'Age': 13, 'Breed': 'Arabian', 'Race Record': '10 wins in 14 races', 'Group': 'B'}, {'Horse ID': 11, 'Horse Name': 'Sparky', 'Jockey Name': 'Peter Edinson', 'Age': 10, 'Breed': 'Thoroughbred', 'Race Record': '7 wins in 10 races', 'Group': 'C'}, {'Horse ID': 12, 'Horse Name': 'Mocha', 'Jockey Name': 'Emma Baker', 'Age': 13, 'Breed': 'Warmbloods', 'Race Record': '9 wins in 15 races', 'Group': 'D'}, {'Horse ID': 13, 'Horse Name': 'Nutmeg', 'Jockey Name': 'Jack Davis', 'Age': 9, 'Breed': 'Gaited Breeds', 'Race Record': '6 wins in 9 races', 'Group': 'C'}, {'Horse ID': 14, 'Horse Name': 'Trixie', 'Jockey Name': 'Helen Ashley', 'Age': 11, 'Breed': 'American Quarter', 'Race Record': '6 wins in 10 races', 'Group': 'D'}, {'Horse ID': 15, 'Horse Name': 'Jackie', 'Jockey Name': 'Louis McKale', 'Age': 10, 'Breed': 'Arabian', 'Race Record': '5 wins in 9 races', 'Group': 'C'}, {'Horse ID': 16, 'Horse Name': 'Hidalgo', 'Jockey Name': 'Carl Peterson', 'Age': 7, 'Breed': 'Draft Breeds', 'Race Record': '3 wins in 5 races', 'Group': 'D'}, {'Horse ID': 17, 'Horse Name': 'Charlie', 'Jockey Name': 'Sara Lennon', 'Age': 10, 'Breed': 'Morgan', 'Race Record': '6 wins in 11 races', 'Group': 'C'}, {'Horse ID': 18, 'Horse Name': 'Billy', 'Jockey Name': 'Hardin Wellington', 'Age': 10, 'Breed': 'Arabian', 'Race Record': '8 wins in 9 races', 'Group': 'D'}, {'Horse ID': 19, 'Horse Name': 'Dash', 'Jockey Name': 'Nick Marley', 'Age': 13, 'Breed': 'Appaloosa', 'Race Record': '10 wins in 13 races', 'Group': 'C'}, {'Horse ID': 20, 'Horse Name': 'Ollie', 'Jockey Name': 'Kate Huston', 'Age': 14, 'Breed': 'American Quarter', 'Race Record': '11 wins in 15 races', 'Group': 'D'}]
```

Figure 5.5: Updating horse details Result

5.2 Code Description

Allows the user to update any of the required horse details.

If the race has already started it will print an error message mentioning the race has started and cannot update records while mentioning cannot update records as a list of details does not exist if the “horse_list” has not been created.

After the user input the “Horse ID” of the record required to be updated, it checks through all the “Horse ID”s with the for loop and allows the user to input each updated detail of the required horse.

As in the code for adding horse details, this code also prints error messages if the user input invalid values for “Horse ID”, “Age” and “Group” and let the user to re-enter the value.

If the user input “Horse ID” cannot be found, it will print that the “Horse ID was not found”.

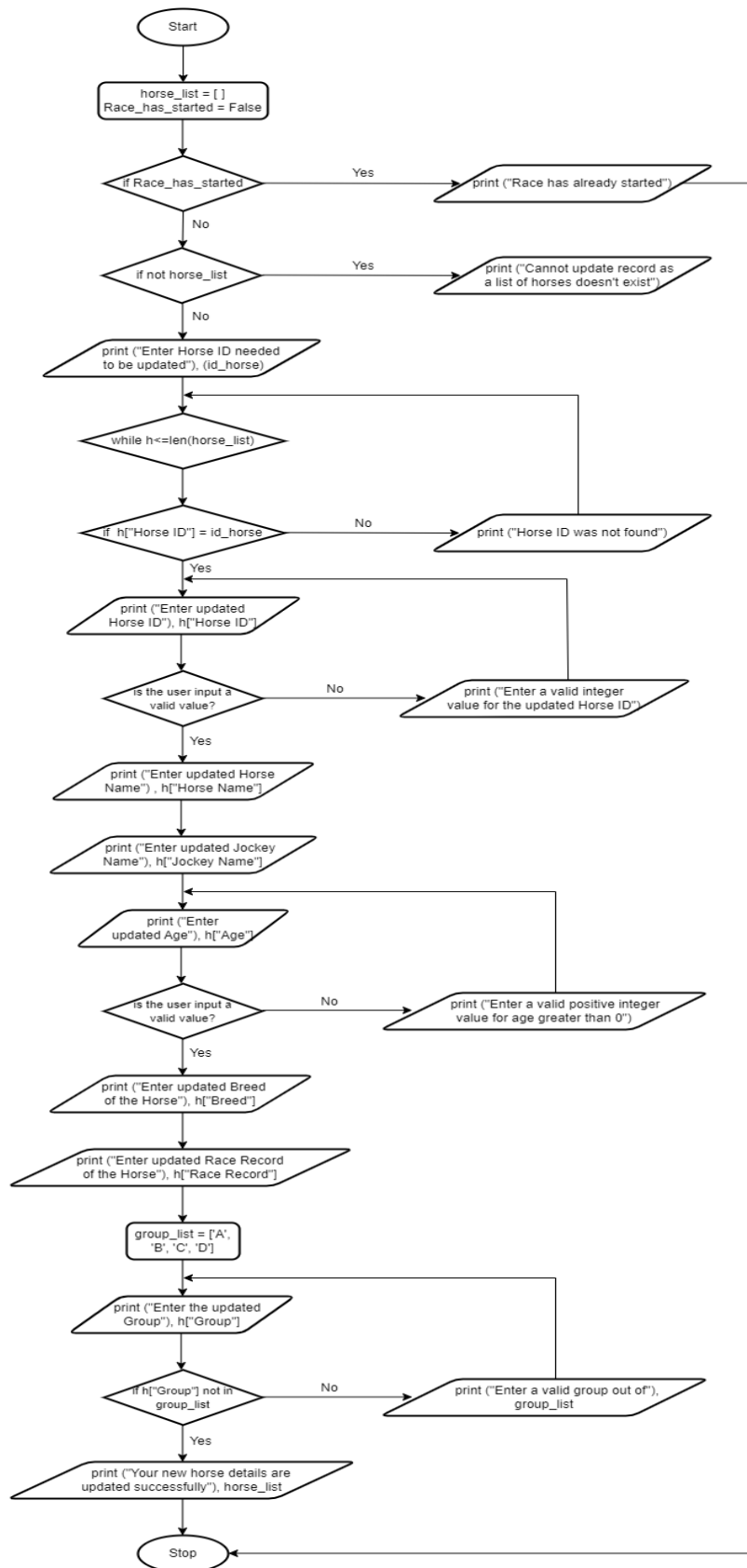


Figure 5.6: Updating horse details Flow Chart

Section 6: View the Registered Horses' Details

6.0 Code

```
Python Coursework.py x
117 # Viewing horse details
118 # Bubble sort
119 1 usage
120 def bubble_sort():
121     global horse_list
122     if not horse_list:
123         print("Cannot execute as a list of horses doesn't exist")
124     else:
125         for i in range(0, len(horse_list) - 1):
126             for j in range(len(horse_list) - 1):
127                 if horse_list[j]["Horse ID"] > horse_list[j + 1]["Horse ID"]:
128                     horse_list[j], horse_list[j + 1] = horse_list[j + 1], horse_list[j]
129
130 1 usage
131 def view_sorted_list():
132     global horse_list
133     if not horse_list:
134         print("Cannot execute as a list of horses doesn't exist")
135     else:
136         bubble_sort()
137         print("\nRegistered horses sorted by horse ID: ")
138         for h in horse_list:
139             print(h)
```

Figure 6.1: Viewing sorted horse details Code

6.1 Result

```
Enter your command: VHD

Registered horses sorted by horse ID:
{'Horse ID': 1, 'Horse Name': 'Maximus', 'Jockey Name': 'David Thompson', 'Age': 10, 'Breed': 'Arabian', 'Race Record': '8 wins in 10 races', 'Group': 'A'}
{'Horse ID': 2, 'Horse Name': 'Ruby', 'Jockey Name': 'Sally Kendall', 'Age': 12, 'Breed': 'Thoroughbred', 'Race Record': '9 wins in 11 races', 'Group': 'A'}
{'Horse ID': 3, 'Horse Name': 'Ranger', 'Jockey Name': 'John Maxwell', 'Age': 8, 'Breed': 'Appaloosa', 'Race Record': '4 wins in 7 races', 'Group': 'A'}
{'Horse ID': 4, 'Horse Name': 'Molly', 'Jockey Name': 'Harry Clandan', 'Age': 13, 'Breed': 'Morgan', 'Race Record': '7 wins in 11 races', 'Group': 'A'}
{'Horse ID': 5, 'Horse Name': 'Joey', 'Jockey Name': 'Belle Kendall', 'Age': 13, 'Breed': 'Warmbloods', 'Race Record': '8 wins in 12 races', 'Group': 'A'}
{'Horse ID': 6, 'Horse Name': 'Fury', 'Jockey Name': 'Daniel Baker', 'Age': 8, 'Breed': 'Gaited Breeds', 'Race Record': '3 wins in 8 races', 'Group': 'B'}
{'Horse ID': 7, 'Horse Name': 'Lucky', 'Jockey Name': 'Jonathan Cooper', 'Age': 15, 'Breed': 'Draft Breeds', 'Race Record': '13 wins in 14 races', 'Group': 'B'}
{'Horse ID': 8, 'Horse Name': 'Dakota', 'Jockey Name': 'Ellie Morgan', 'Age': 8, 'Breed': 'American Quarter', 'Race Record': '5 wins in 10 races', 'Group': 'B'}
{'Horse ID': 9, 'Horse Name': 'Chad', 'Jockey Name': 'Edward Matthew', 'Age': 10, 'Breed': 'Appaloosa', 'Race Record': '7 wins in 12 races', 'Group': 'B'}
{'Horse ID': 10, 'Horse Name': 'Dusty', 'Jockey Name': 'Henry Kelson', 'Age': 13, 'Breed': 'Arabian', 'Race Record': '10 wins in 14 races', 'Group': 'B'}
{'Horse ID': 11, 'Horse Name': 'Sparky', 'Jockey Name': 'Peter Edinson', 'Age': 10, 'Breed': 'Thoroughbred', 'Race Record': '7 wins in 10 races', 'Group': 'C'}
{'Horse ID': 12, 'Horse Name': 'Mocha', 'Jockey Name': 'Emma Baker', 'Age': 13, 'Breed': 'Warmbloods', 'Race Record': '9 wins in 15 races', 'Group': 'D'}
{'Horse ID': 13, 'Horse Name': 'Nutmeg', 'Jockey Name': 'Jack Davis', 'Age': 9, 'Breed': 'Gaited Breeds', 'Race Record': '6 wins in 9 races', 'Group': 'C'}
{'Horse ID': 14, 'Horse Name': 'Trixie', 'Jockey Name': 'Helen Ashley', 'Age': 11, 'Breed': 'American Quarter', 'Race Record': '6 wins in 10 races', 'Group': 'D'}
{'Horse ID': 15, 'Horse Name': 'Jackie', 'Jockey Name': 'Louis McKale', 'Age': 10, 'Breed': 'Arabian', 'Race Record': '5 wins in 9 races', 'Group': 'C'}
{'Horse ID': 16, 'Horse Name': 'Hidalgo', 'Jockey Name': 'Carl Peterson', 'Age': 7, 'Breed': 'Draft Breeds', 'Race Record': '3 wins in 5 races', 'Group': 'D'}
```

Figure 6.2: Viewing sorted horse details Result

```
{'Horse ID': 17, 'Horse Name': 'Charlie', 'Jockey Name': 'Sara Lennon', 'Age': 10, 'Breed': 'Morgan', 'Race Record': '6 wins in 11 races', 'Group': 'C'}  
{'Horse ID': 18, 'Horse Name': 'Billy', 'Jockey Name': 'Hardin Wellington', 'Age': 10, 'Breed': 'Arabian', 'Race Record': '8 wins in 9 races', 'Group': 'D'}  
{'Horse ID': 19, 'Horse Name': 'Dash', 'Jockey Name': 'Nick Marley', 'Age': 13, 'Breed': 'Appaloosa', 'Race Record': '10 wins in 13 races', 'Group': 'C'}  
{'Horse ID': 20, 'Horse Name': 'Ollie', 'Jockey Name': 'Kate Huston', 'Age': 14, 'Breed': 'American Quarter', 'Race Record': '11 wins in 15 races', 'Group':  
'D'}
```

Figure 6.3: Viewing sorted horse details Result

6.2 Code Description

Allows the user to view the horse details sorted according to the “Horse ID”.

If the list “horse_list” has not been created it will show that the function cannot be executed as the list is not available.

To sort the horse details according to the “Horse ID”, the code has used “Bubble Sort” function and for loop inside the code.

After executing the sorting part, it will call the “Bubble Sort” function and the required sorted list will be printed after that.

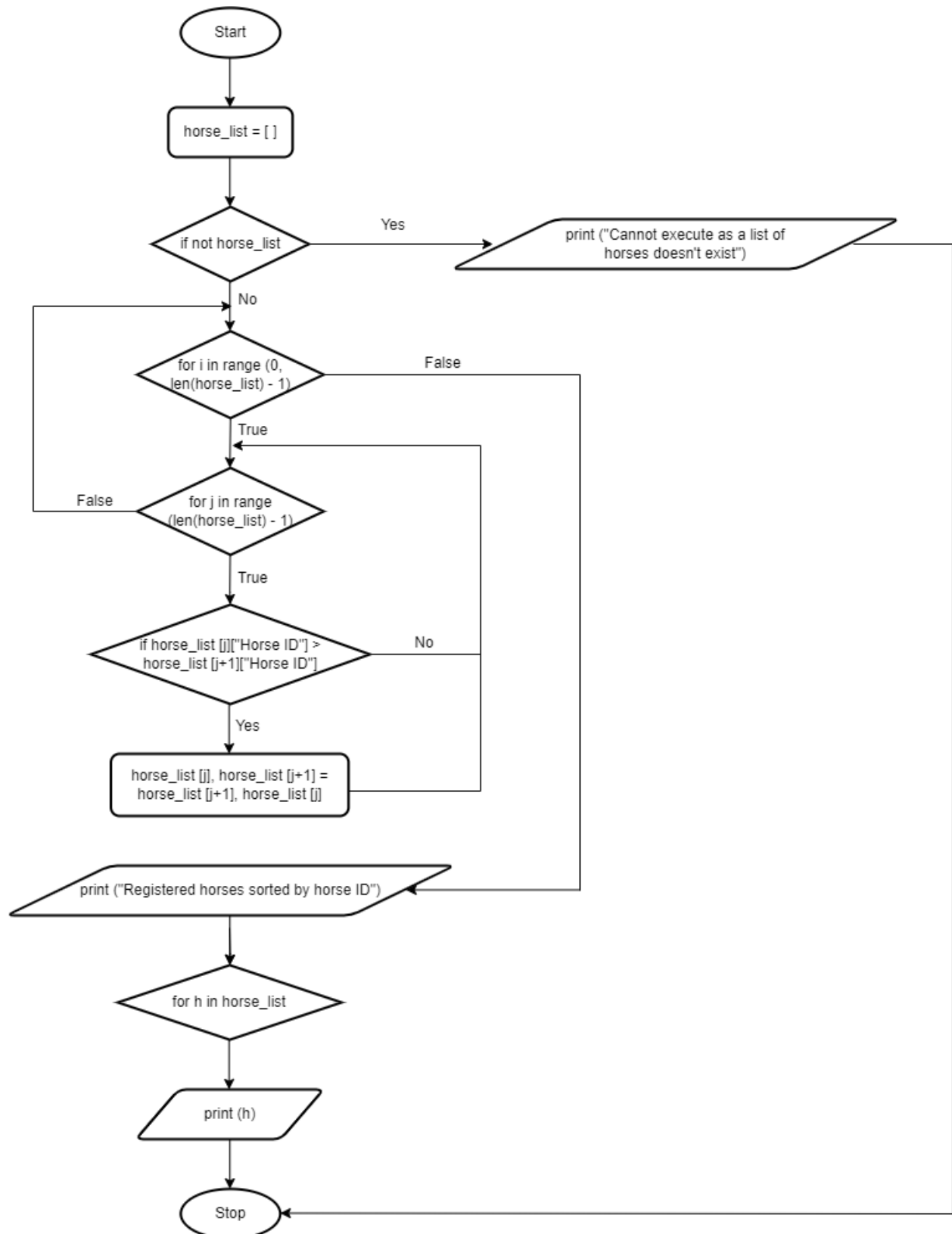


Figure 6.4: Viewing sorted horse details Flow Chart

Section 7: Save the Horse Details to the Text File

7.0 Code

```
Python Coursework.py ×  
141 # Saving the details into a file  
142 # Arranging horse list according to the groups  
143 1 usage  
143 def save_to_text_file():  
144     global horse_list  
145     group_horse = {}  
146     if not horse_list:  
147         print("Cannot execute as a list of horses doesn't exist")  
148     else:  
149         for h in horse_list:  
150             group = h["Group"]  
151             if group not in group_horse:  
152                 group_horse[group] = []  
153             group_horse[group].append(h)
```

Figure 7.1: Saving horse details Code

```
Python Coursework.py ×  
152         group_horse[group].append(h)  
153  
154     # Saving into the file  
155     f = open(file_path, 'w')  
156     for group, g in group_horse.items():  
157         f.write(f"\nGroup{group}:\n")  
158         for h in g:  
159             f.write(str(h) + "\n")  
160     f.close()  
161  
162     fr = open(file_path, 'r')  
163     print(fr.read())  
164     fr.close()  
165     print("Horse details have been saved to the file successfully!")
```

Figure 7.2: Saving horse details Code

7.1 Result

Enter your command: *SHD*

```
GroupA:
{'Horse ID': 1, 'Horse Name': 'Maximus', 'Jockey Name': 'David Thompson', 'Age': 10, 'Breed': 'Arabian', 'Race Record': '8 wins in 10 races', 'Group': 'A'}
{'Horse ID': 2, 'Horse Name': 'Ruby', 'Jockey Name': 'Sally Kendall', 'Age': 12, 'Breed': 'Thoroughbred', 'Race Record': '9 wins in 11 races', 'Group': 'A'}
{'Horse ID': 3, 'Horse Name': 'Ranger', 'Jockey Name': 'John Maxwell', 'Age': 8, 'Breed': 'Appaloosa', 'Race Record': '4 wins in 7 races', 'Group': 'A'}
{'Horse ID': 4, 'Horse Name': 'Molly', 'Jockey Name': 'Harry Clanden', 'Age': 13, 'Breed': 'Morgan', 'Race Record': '7 wins in 11 races', 'Group': 'A'}
{'Horse ID': 5, 'Horse Name': 'Joey', 'Jockey Name': 'Belle Kendall', 'Age': 13, 'Breed': 'Warmbloods', 'Race Record': '8 wins in 12 races', 'Group': 'A'}

GroupB:
{'Horse ID': 6, 'Horse Name': 'Fury', 'Jockey Name': 'Daniel Baker', 'Age': 8, 'Breed': 'Gaited Breeds', 'Race Record': '3 wins in 8 races', 'Group': 'B'}
{'Horse ID': 7, 'Horse Name': 'Lucky', 'Jockey Name': 'Jonathan Cooper', 'Age': 15, 'Breed': 'Draft Breeds', 'Race Record': '13 wins in 14 races', 'Group': 'B'}
{'Horse ID': 8, 'Horse Name': 'Dakota', 'Jockey Name': 'Ellie Morgan', 'Age': 8, 'Breed': 'American Quarter', 'Race Record': '5 wins in 10 races', 'Group': 'B'}
{'Horse ID': 9, 'Horse Name': 'Chad', 'Jockey Name': 'Edward Matthew', 'Age': 10, 'Breed': 'Appaloosa', 'Race Record': '7 wins in 12 races', 'Group': 'B'}
{'Horse ID': 10, 'Horse Name': 'Dusty', 'Jockey Name': 'Henry Kelson', 'Age': 13, 'Breed': 'Arabian', 'Race Record': '10 wins in 14 races', 'Group': 'B'}
```

Figure 7.3: Saving horse details Result

```
GroupC:
{'Horse ID': 11, 'Horse Name': 'Sparky', 'Jockey Name': 'Peter Edinson', 'Age': 10, 'Breed': 'Thoroughbred', 'Race Record': '7 wins in 10 races', 'Group': 'C'}
{'Horse ID': 13, 'Horse Name': 'Nutmeg', 'Jockey Name': 'Jack Davis', 'Age': 9, 'Breed': 'Gaited Breeds', 'Race Record': '6 wins in 9 races', 'Group': 'C'}
{'Horse ID': 15, 'Horse Name': 'Jackie', 'Jockey Name': 'Louis McKale', 'Age': 10, 'Breed': 'Arabian', 'Race Record': '5 wins in 9 races', 'Group': 'C'}
{'Horse ID': 17, 'Horse Name': 'Charlie', 'Jockey Name': 'Sara Lennon', 'Age': 10, 'Breed': 'Morgan', 'Race Record': '6 wins in 11 races', 'Group': 'C'}
{'Horse ID': 19, 'Horse Name': 'Dash', 'Jockey Name': 'Nick Marley', 'Age': 13, 'Breed': 'Appaloosa', 'Race Record': '10 wins in 13 races', 'Group': 'C'}

GroupD:
{'Horse ID': 12, 'Horse Name': 'Mocha', 'Jockey Name': 'Emma Baker', 'Age': 13, 'Breed': 'Warmbloods', 'Race Record': '9 wins in 15 races', 'Group': 'D'}
{'Horse ID': 14, 'Horse Name': 'Trixie', 'Jockey Name': 'Helen Ashley', 'Age': 11, 'Breed': 'American Quarter', 'Race Record': '6 wins in 10 races', 'Group': 'D'}
{'Horse ID': 16, 'Horse Name': 'Hidalgo', 'Jockey Name': 'Carl Peterson', 'Age': 7, 'Breed': 'Draft Breeds', 'Race Record': '3 wins in 5 races', 'Group': 'D'}
{'Horse ID': 18, 'Horse Name': 'Billy', 'Jockey Name': 'Hardin Wellington', 'Age': 10, 'Breed': 'Arabian', 'Race Record': '8 wins in 9 races', 'Group': 'D'}
{'Horse ID': 20, 'Horse Name': 'Ollie', 'Jockey Name': 'Kate Huston', 'Age': 14, 'Breed': 'American Quarter', 'Race Record': '11 wins in 15 races', 'Group': 'D'}
```

Figure 7.4: Saving horse details Result

7.2 Code Description

Allows the user to save the horse details to the text file according to the four groups.

If the list “horse_list” has not been created it will show that the function cannot be executed as the list is not available.

In the first part of the code, it will categorize the horse details according to the groups using the for loop.

In the second part, it will open the text file “horse_details.txt” in ‘w’ mode and write the categorized details to the file.

After that it will reopen the text file in the ‘r’ mode and let the user see the saved details and print that the details have been saved to the file.

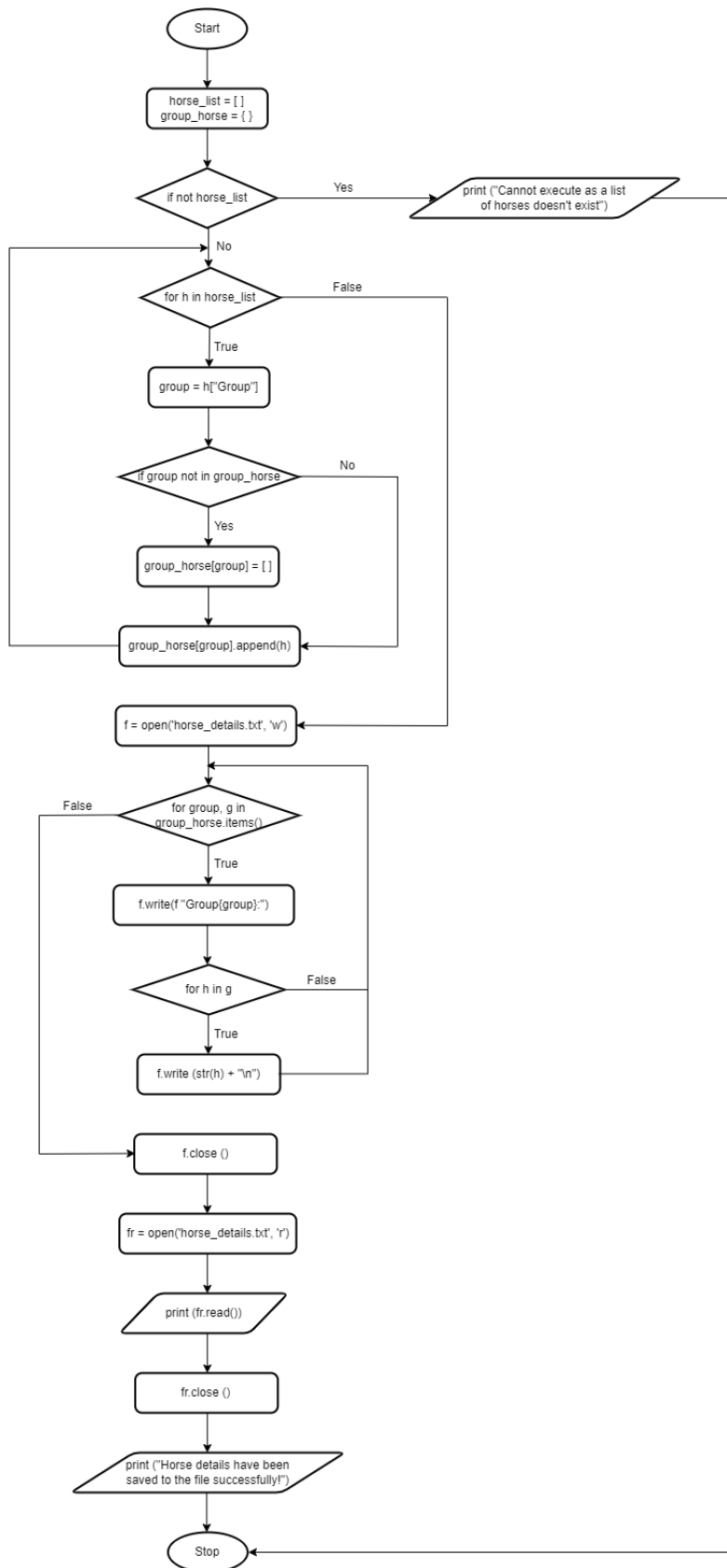


Figure 7.5: Saving horse details Flow Chart

Section 8: Select Four Horses Randomly

8.0 Code

Python Coursework.py ×

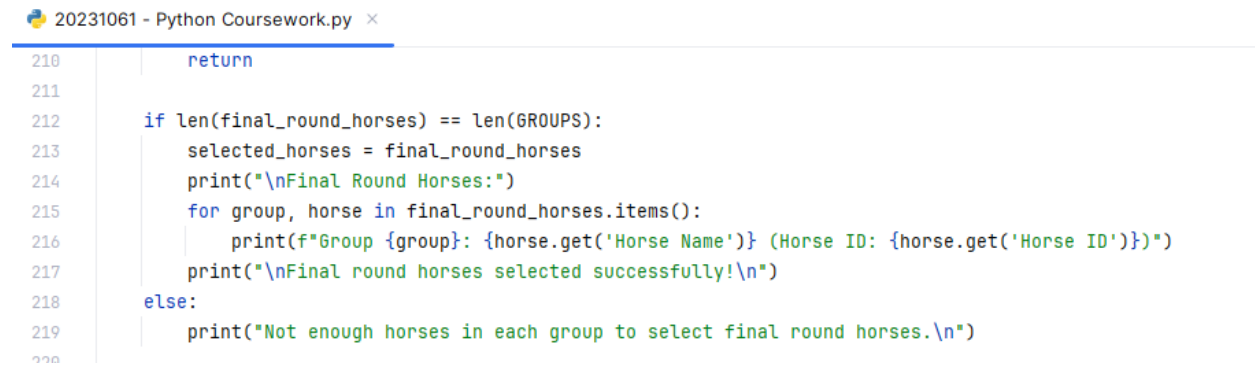
```
168 # Function to select horses
169 1 usage
169 def selection_for_major_round():
170     global horse_list, selected_horses, file_path
171
172     try:
173         if not horse_list:
174             print("No horses registered. Cannot select.\n")
175         else:
176             file = open(file_path, 'r')
177             key = None # Initialize key outside the loop
178             value = None # Initialize value outside the loop
179             for line in file:
180                 line = line.strip()
181                 if line:
182                     parts = line.split()
183                     if len(parts) >= 2:
184                         key, value = parts[0], ' '.join(parts[1:])
185                         key = key.strip()
186                         value = value.strip()
187                         if key == 'Group':
188                             horse_list.append({'Group': value})
189                         else:
190                             horse_list[-1][key] = value
191             file.close()
```

Figure 8.1: Selecting horses Code

20231061 - Python Coursework.py ×

```
191         file.close()
192     except ValueError as ve:
193         print(f"Error adding horse details: {ve}\n")
194     except FileNotFoundError:
195         print(f"Error: File not found at {file_path}\n")
196     except Exception as e:
197         print(f"Error loading horse details: {e}\n")
198     return
199
200 final_round_horses = {}
201
202 try:
203     for group in GROUPS:
204         group_horses = [horse for horse in horse_list if horse.get('Group') == group]
205         if group_horses:
206             selected_horse = random.choice(group_horses)
207             final_round_horses[group] = selected_horse
208     except Exception as e:
209         print(f"Error selecting final round horses: {e}\n")
210     return
```

Figure 8.2: Selecting horses Code



```

20231061 - Python Coursework.py x
210     return
211
212     if len(final_round_horses) == len(GROUPS):
213         selected_horses = final_round_horses
214         print("\nFinal Round Horses:")
215         for group, horse in final_round_horses.items():
216             print(f"Group {group}: {horse.get('Horse Name')} (Horse ID: {horse.get('Horse ID')})")
217         print("\nFinal round horses selected successfully!\n")
218     else:
219         print("Not enough horses in each group to select final round horses.\n")

```

Figure 8.3: Selecting horses Code

8.1 Result

Enter your command: `SDD`

Final Round Horses:

Group A: Ranger (Horse ID: 3)

Group B: Fury (Horse ID: 6)

Group C: Dash (Horse ID: 19)

Group D: Hidalgo (Horse ID: 16)

Final round horses selected successfully!

Figure 8.4: Selecting horses Result

8.2 Code Description

This code selects four random horses from the four groups. If the list “horse_list” has not been created it will show that the function cannot be executed as the list is not available.

In the first part of the code, it will open the “horse_details.txt” file, reads the content and append a dictionary with the group information to the “horse_list” if the line represents a “Group”. This also handles different exceptions occurred during file reading.

In the second part, it will initialize “final_round_horses” dictionary, iterates through predefined “GROUPS”, select random horses from the four groups and add them to the dictionary.

In the last part it will print the final round horses. If not enough horses are available in each group, it prints a message indicating insufficient horses.

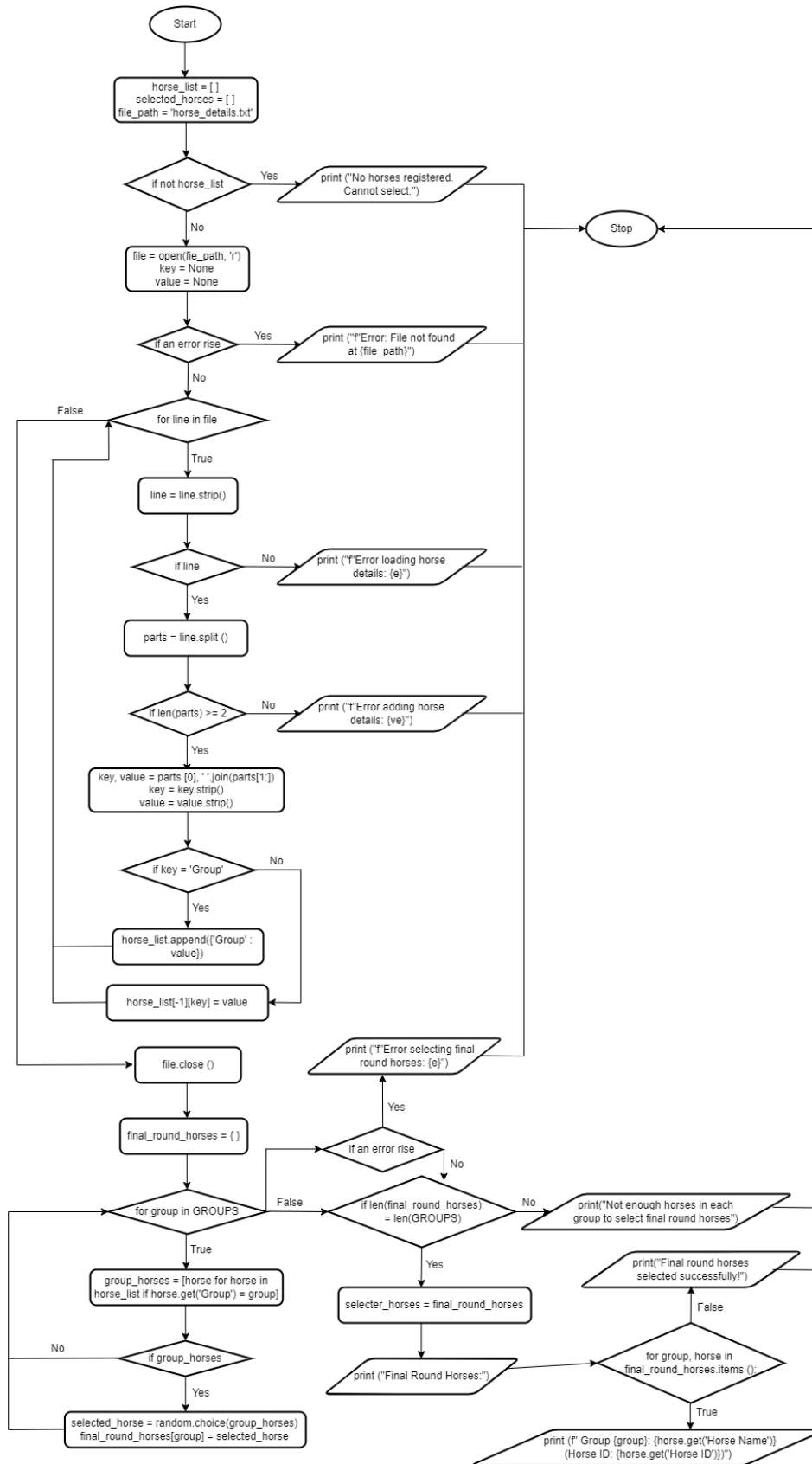


Figure 8.5: Selecting horses Flow Chart

Section 9: Display Winning Horses

9.0 Code

```
Python Coursework.py x
223
224 # Function to display winning horse details
225 1 usage
226 def display_winners():
227     global selected_horses, race_times
228
229     if selected_horses:
230         race_times = {}
231
232         for group, horse in selected_horses.items():
233             race_time = random.randint(a=10, b=90)
234             race_times[horse['Horse ID']] = race_time
235
236         print("\nRace Times for Final Round Horses:")
237         for horse_id in race_times:
238             print(f"Horse ID: {horse_id}, Time: {race_times[horse_id]} seconds")
239
240         sorted_horses = list(race_times.items())
241         n = len(sorted_horses)
242         for i in range(n - 1):
243             for j in range(0, n - i - 1):
244                 if sorted_horses[j][1] > sorted_horses[j + 1][1]:
245                     sorted_horses[j], sorted_horses[j + 1] = sorted_horses[j + 1], sorted_horses[j]
```

Figure 9.1: Display winning horses Code

```
Python Coursework.py x
246 print("\nWinners:")
247 for i in range(min(3, len(sorted_horses))):
248     horse_id, time = sorted_horses[i]
249     horse = None
250     for h in selected_horses.values():
251         if h['Horse ID'] == horse_id:
252             horse = h
253             break
254     if horse:
255         print(f"{i + 1} Place: {horse['Horse Name']} (Horse ID: {horse['Horse ID']}), Time: {time} seconds")
256
257 else:
258     print("No final round horses selected. Please run 'SDD' to select horses for the final round.\n")
259
```

Figure 9.2: Display winning horses Code

9.1 Result

Enter your command: *WHD*

Race Times for Final Round Horses:

Horse ID: 3, Time: 86 seconds

Horse ID: 6, Time: 50 seconds

Horse ID: 19, Time: 55 seconds

Horse ID: 16, Time: 66 seconds

Winners:

1 Place: Fury (Horse ID: 6), Time: 50 seconds

2 Place: Dash (Horse ID: 19), Time: 55 seconds

3 Place: Hidalgo (Horse ID: 16), Time: 66 seconds

Figure 9.3: Display winning horses Result

9.2 Code Description

This code prints the details of the horses which won the race.

If no final round horses are selected it will not execute the code and prints a message asking the user to run the selection “SDD” code.

For each horse selected to the final round, it will assign a random time between 10-90 seconds and store in the “race_times” dictionary.

Using the “Bubble Sort” function it will sort the horses based on their race time in the ascending order and create the “sorted_horses” list.

After printing the race time, it iterates through the “sorted_horses” using the for loop and print the top three horses.

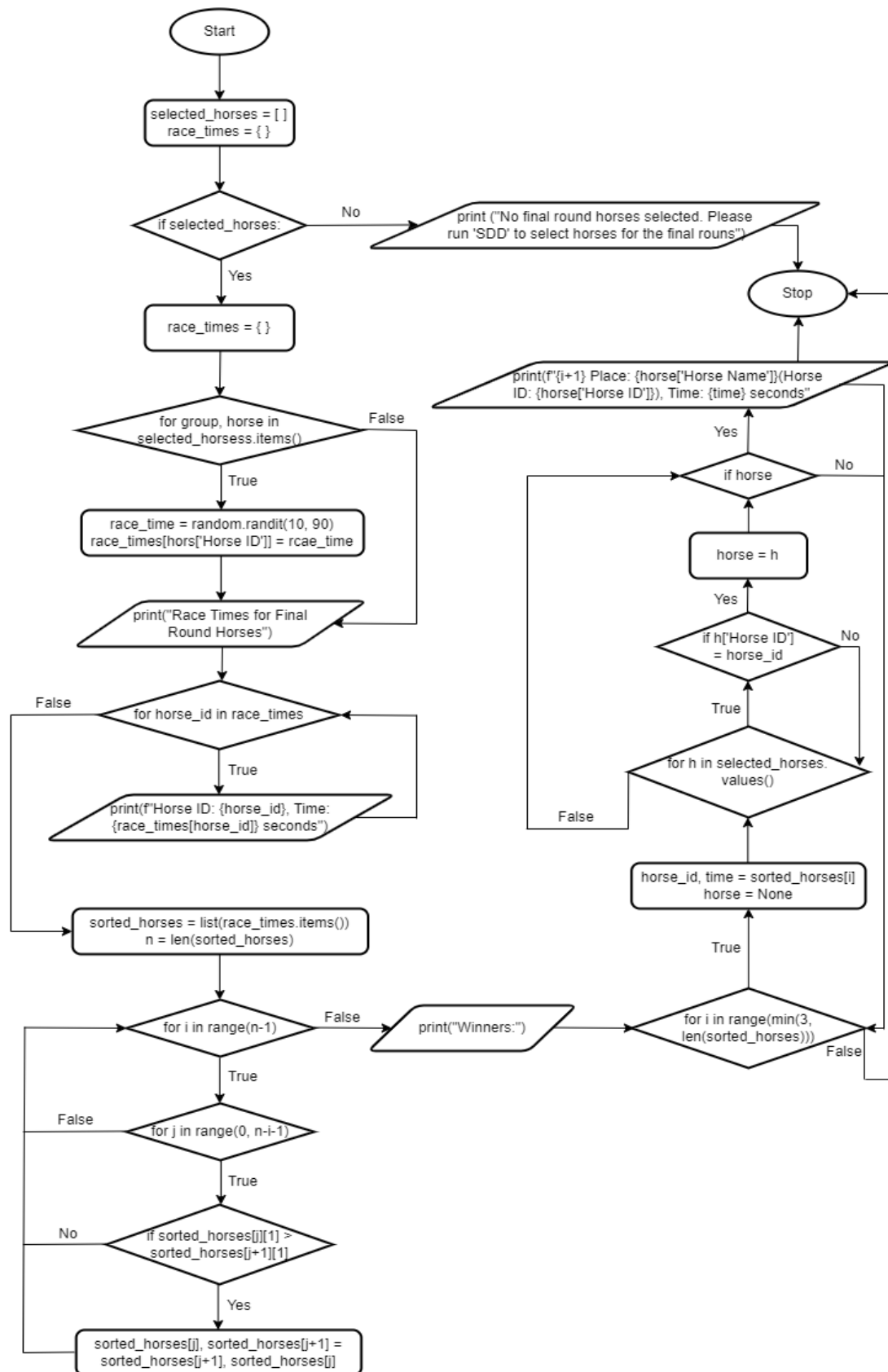


Figure 9.4: Display winning horses Flow Chart

Section 10: Visualize Winning Horses

10.0 Code

```
Python Coursework.py x
260
261 # Function to visualize winning horses
262 1 usage
263 def visualizing_winner(race_times):
264     global selected_horses
265
266     if selected_horses and race_times:
267         print("\nVisualization of Time Spent by Winning Horses:")
268
269         sorted_horses = list(race_times.items())
270
271         n = len(sorted_horses)
272         for i in range(n):
273             for j in range(0, n-i-1):
274                 if sorted_horses[j][1] > sorted_horses[j+1][1]:
275                     sorted_horses[j], sorted_horses[j+1] = sorted_horses[j+1], sorted_horses[j]
276
277         i = 0
278         while i < 3 and i < len(sorted_horses):
279             horse_id, time = sorted_horses[i]
280             horse = None
281             for h in selected_horses.values():
282                 if h['Horse ID'] == horse_id:
283                     horse = h
284                     break
285             if horse:
286                 stars = '*' * (time // 10)
```

Figure 10.1: Visualize winning horses Code

```
Python Coursework.py x
284
285     if horse:
286         stars = '*' * (time // 10)
287         print(f"{horse['Horse Name']}: {stars:<20} {time}s ({i + 1}st Place)")
288         i += 1
289
290     else:
291         print("No final round horses selected. Please run 'SDD' to select horses for the final round.\n")
```

Figure 10.2: Visualize winning horses Code

10.1 Result

Enter your command: `VWH`

Visualization of Time Spent by Winning Horses:

Fury:	*****	50s (1st Place)
Dash:	*****	55s (2st Place)
Hidalgo:	*****	60s (3st Place)

Figure 10.3: Visualize winning horses Result

10.2 Code Description

This code visualizes the winning horses using “*” sign.

If no final round horses are selected it will not execute the code and prints a message asking the user to run the selection “SDD” code.

If there are selected horses and race times, it will sort the horses based on their race times using the “Bubble Sort” function. It iterates through the “sorted_horses” using while loop and calculate the number of “*” based on the race times, with each “*” representing 10 seconds.

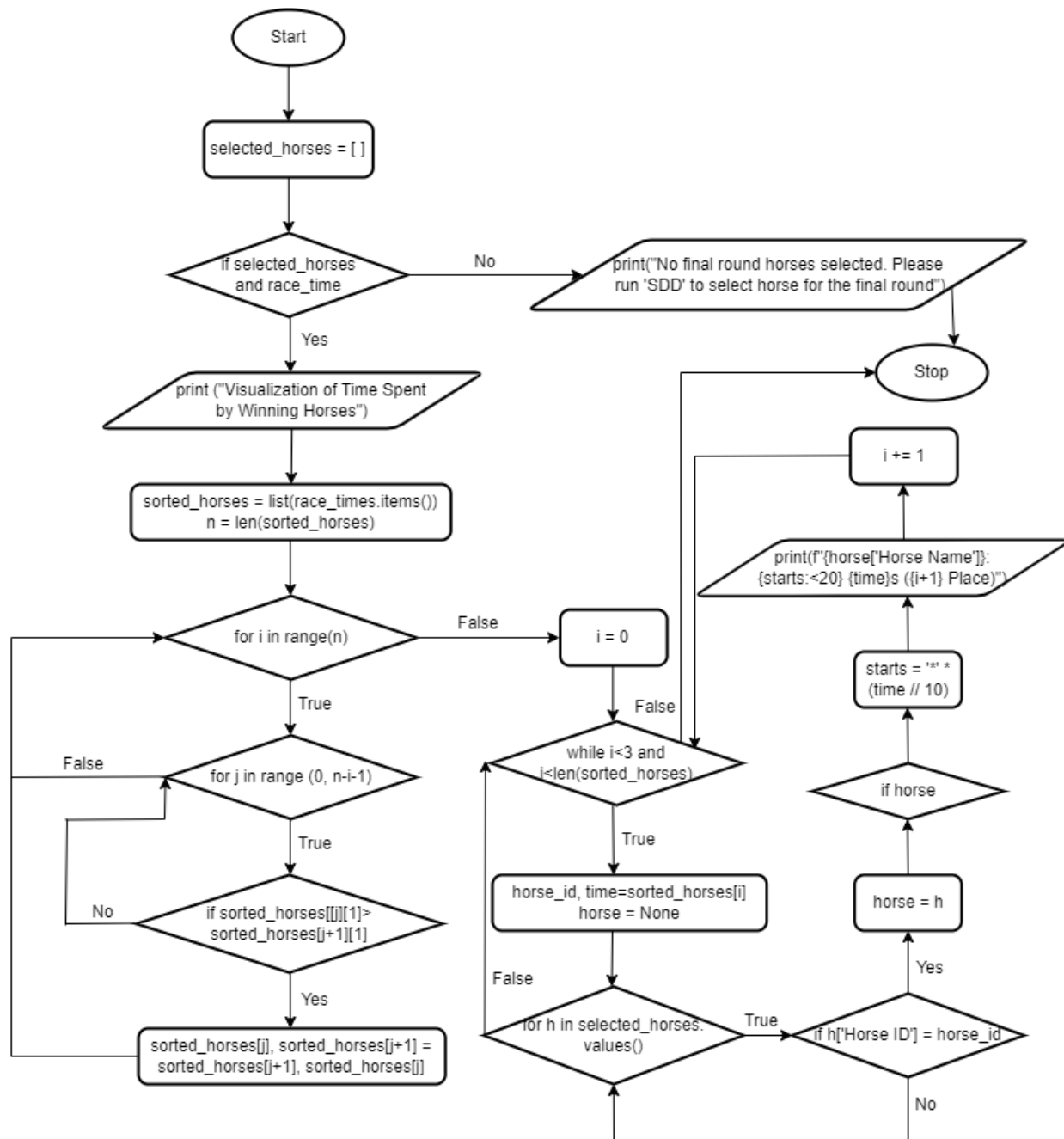


Figure 10.4: Visualize winning horses Flow Chart

Section 11: Exit Program

11.0 Code

```
Enter your command: ESC  
Exit
```

```
Process finished with exit code 0
```

Figure 11.1: Exit program Code

11.1 Code Description

Allows user to end the program.

After entering “ESC” the program will be terminated.

References

Refsnes Data (2023). *Python Tutorial*. Available at: <https://www.w3schools.com/python/> (Accessed 15 December 2023)

GeeksforGeeks (2023). *Python Tutorial*. Available at: <https://www.geeksforgeeks.org/python-programming-language/?ref=outind/> (Accessed 17 December 2023)